Computer Communications 154 (2020) 433–441

Contents lists available at ScienceDirect

Computer Communications

journal homepage: www.elsevier.com/locate/comcom

A decision based unsymmetrical trimmed modified winsorized variants for the removal of high density salt and pepper noise in images and videos

Vasanth kishorebabu [a,*], R. Varatharajan [b]

[a] Department of E.C.E, Vidya Jyothi Institute of Technology, Aziz Nagar Gate, chilukur road, Hyderabad-78, Telengana, India
[b] Department of Electronics and Communication Engineering, School of Electronics Engineering Bharath University, Chennai, India

ARTICLE INFO

Keywords:
Unsymmetrical trimmed modified winsorized variants
Salt and pepper noise
Edge preservation

ABSTRACT

A Decision Based Unsymmetrical trimmed modified Winsorized Variants for the removal of high density salt and pepper noise in images and videos is proposed. The algorithm initially checks for fixed outliers (0 or 255) in the images in a fixed 3x3 neighbourhood. If the processed pixel is an outlier then 4 neighbours are checked for noise. If all 4 neighbours are noisy then mean of 4 neighbours replaces the corrupted pixel. If 4 neighbours are not noisy then number of non noisy pixels is used to replace the corrupted pixel either with unsymmetrical trimmed modified Winsorized median or midpoint. The processed pixel is left unaltered if the pixel does not hold the outliers. The exhaustive experiments on standard database images and videos indicate that the DBUTMWV algorithm has good noise elimination capabilities with excellent information preservation characteristics even at very high noise densities.

1. Introduction

Images and Videos are often damaged with fixed valued impulse noise (salt and pepper noise) because of poor communication channels or A/D Conversion. Filters were the obvious choice for the noise removal. Linear filters were good in eliminating outliers but blurred the image. Non linear filters such as median filters eliminated the outliers by preserving the information content of the image. A standard median filter of fixed kernel was found good at noise elimination but flatters at high noise densities. A larger window removes the noise but makes the image information less. Hence the size of the window should be carefully chosen [1]. An adaptive window will remove the noise but blurred the image [2]. A special class of filters such as weighted median filter [3], centre weighted median filters [4] and Tristate filters [5] were used to eliminate the outlier. All the above algorithms replaced the pixels of an image with a value irrespective the pixel value is noisy or not. Progressive switched median filter (PSMF) checked for the noisy pixel using a detector and corrected with median filter [6]. The filter performance of the filter diminishes at high noise densities. Filtering was done using robust estimators were introduced [7]. A novel adaptive switched filter, [8] which used fuzzy set for noise removal. A two level scheme [9] was introduced that used adaptive median filter for corrupted pixel identification followed by regularization method for pixels that are identified as corrupted. Decision based median filter (DMF) [10] used pre-processed neighbourhood pixels if the computed median is noisy. This induced streaks in images. The streaks was minimized by an improved version of median filter (IDMF) [11], which replaced mean of pre-processed neighbours. Cascaded algorithms [12] used decision based filters initially followed by trimmed mean or midpoint. These algorithms smeared the edges at high noise densities. A new class of unsymmetrical trimmed filters were introduced that used median [12], mean [13], midpoint [14] for high density outlier removal. These algorithms exhibited fading effect. Interpolation techniques such as continued fractions [15], Spline [16], radial bias function [17], Kriging [18], inverse distance weighted [19] were employed for the elimination of outliers. These algorithms require adaptive window or complex calculations for evaluation of restored pixel. Hence a suitable algorithm has to be formulated such that it detects and correct the noisy pixels at very high noise densities by preserving the information content of the images and videos. Section 2 gives the flowchart of the DBUTMWV algorithm. Section 3 deals with the insight of the methodology used. Section 4 gives the simulation results and discussions made. Section 5 concludes the paper.

2. Decision based Unsymmetrical Trimmed Modified Winsorized Variants (DBUTMWV)

The algorithm Decision based Unsymmetrical Trimmed Modified Winsorized Variants (DBUTMWV) will be referred as DBUTMWV algorithm thereafter. The algorithm uses a fixed 3 × 3 window for increasing noise densities. The algorithm initially checks the 4 neighbours for outliers. If the neighbours are noisy then mean of the 4 neighbours. If anyone neighbours are not noisy then number of noisy

* Corresponding author.
E-mail addresses: vasanthecek@gmail.com (K. Vasanth), varathu21@yahoo.com (R. Varatharajan).

https://doi.org/10.1016/j.comcom.2020.02.048
Received 7 July 2019; Received in revised form 21 January 2020; Accepted 17 February 2020
Available online 4 March 2020
0140-3664/© 2020 Elsevier B.V. All rights reserved.

Abbrevations

SPN	Salt and pepper noise
ADKIF	Adaptive Decision based Kriging Interpolation Filter
PSNR	Peak Signal to noise ratio
MSE	Mean Square Error
IEF	Image Enhancement Factor
ER	Error Rate
SSIM	Structural Similarity Index metric
FOM	Figure of merit
SMF	Standard Median Filter
AMF	Adaptive Median Filter
CWF	Center weighted Filter
αTMF	Alpha trimmed mean filter
PSMF	Progressive switched median filter
MDBMF	Modified Decision based Median filter
DBA	Decision Based Median algorithm
IDBA	Improved Decision based Median Filter
CUTMF	Cascaded unsymmetrical trimmed mean filter
CUDBMPF	Cascaded Unsymmetrical decision based Midpoint filter
MDBUTMF	Modified Decision based Unsymmetrical Trimmed median filter
MDBUTMF_GM	Modified Decision based Unsymmetrical Trimmed median filter with global mean
ACBSA	Adaptive Cardinal B Spline interpolation filter
NAFSM	Noise adaptive fuzzy switched median
AWMF	Adaptive weighted mean filter
CPBD	Cumulative probability blur detection Metric
NCC	Normalized Correlation Coefficient
DBUTMWV	Decision based Unsymmetrical Trimmed Modified Winsorized Variants

pixels is computed. If the count is less than three then the corrupted pixel is replaced with unsymmetrical trimmed modified Winsorized median else the corrupted pixel is replaced with unsymmetrical trimmed modified Winsorized midpoint. The Fig. 1 depicts the flowchart of the DBUTMWV algorithm on images. The above process is applied on the entire image. In case of videos, each frame are applied with the DBUTMWV algorithm, and later concatenated. If the entire pixels of the window are noisy then the pixels are left unaltered considering it as constant background.

2.1. Unsymmetrical trimmed modified winsorized statistics

The unsymmetrical trimmed Modified Winsorized statistics is a small modification of existing Winsorized operation. The later eliminates a fixed number of samples and replaces the first and last elements of the samples that is retained with number of samples eliminated. In the unsymmetrical trimmed modified Winsorized operation does not eliminate the fixed number of samples on either sides instead outliers are eliminated asymmetrically based on the number present in the array. There are four types of Unsymmetrical Trimmed Modified Winsorized statistics. Unsymmetrical trimmed modified Winsorized mean, median, midpoint and Mode.

Initially, unsymmetrical trimmed modified Winsorized operation is calculated later the trimmed array is subjected to mean, median, mode and midpoint. The above gives the different unsymmetrical trimmed modified Winsorized statistics.

2.2. How the statistic is good in outlier removal?

The DBUTMWV statistics falls well within Arithmetic mean and median and precisely very close to median. This enables the statistic good in eliminating outliers. Statistics such as Unsymmetrical trimmed modified Winsorized mean, median and midpoint was found good in eliminating high density outliers with information preservation but the mode was not found good in removing the noise.

2.3. Why algorithm uses 3 noisy pixels as deciding factor for usage of two different statistics?

A minimum of three noisy pixels is an indication that the number of non noisy pixels is around 67% of the sample. Hence it is good for unsymmetrical trimmed modified Winsorized median to be computed, also this statistics preserve information content of the image. If the number of pixels is greater than 3 then the percentage of non noisy pixels reduces. This is suitable for unsymmetrical trimmed modified Winsorized midpoint. These values of computed median and midpoint lie very close to trimmed mean and median. Hence 3 noisy pixels act as deciding factor for the usage of two different statistics.

3. Insight on DBUTMWV

The insight of the DBUTMWV algorithm is given in Section 3. Each condition is illustrated with a separate example. A Rectangular box indicates the processed window. A circle within it indicates the processed pixel. The bigger matrix is considered as a Image Segment over which the fixed kernel moves

Case(1): The processed Pixel is noisy (0) also all pixels of the window is noisy

In Case 1 the computed pixel is noisy (0). This pixel is considered to be an outlier as it holds 0, also all the elements of the processed window is noisy. Hence the pixels are considered to have uniform pixels around the confined neighbourhood. Hence the processed pixel is retained as shown in Fig. 2.

Case(2) : The processed Pixel is noisy (255) also the 4 neighbours are also noisy

In case 2 the processed pixel is noisy as it holds 255. The 4 neighbours 0, 255, 0, 255 are considered to be noisy. Hence mean of the 4 neighbours is found as follows $(0+255+0+255)/ 4 = 128$. The computed value replaces the corrupted pixel as illustrated in Fig. 3.

Case(3) : The processed Pixel is noisy (255), some of the 4 neighbours are not noisy also number of non noisy pixel inside the processed pixel is greater than 3

In Case 3 the processed pixel hold 0 which is considered as noisy also not all 4 neighbours are noisy. Few pixels of 4 neighbours such as 121 and 124 is considered as non noisy. Convert 2D data into 1D data. Sort the data in increasing order. Eliminate 0 and 255 from the sorted array. Find the Modified Winsorized Mean of the trimmed array. Replace the corrupted pixel with the modified Winsorized midpoint as illustrated in Fig. 4.

Modified Winsorized Midpoint:

Replicate the minimum value and maximum value of the trimmed array and find the mean of the resultant array.

Unsorted array: 123 255 123 255 0 124 123 121 122

Sorted Array: 0 121 122 123 123 123 124 255 255

Trimmed array: 121 122 123 123 123 124

Modified Winsorized Midpoint of trimmed array: (121 121 122 123 123 123 124 124) = (121 + 124)/2 = 122.5 = 123.

Case(4) : The processed Pixel is noisy (255), some of the 4 neighbours are not noisy also number of noisy pixel inside the processed pixel is less than 3

In the case 4 the processed pixel hold 255 which is considered to be an outlier also not all 4 neighbours are noisy. Count the number of noisy pixel inside the current processed window. In this case the

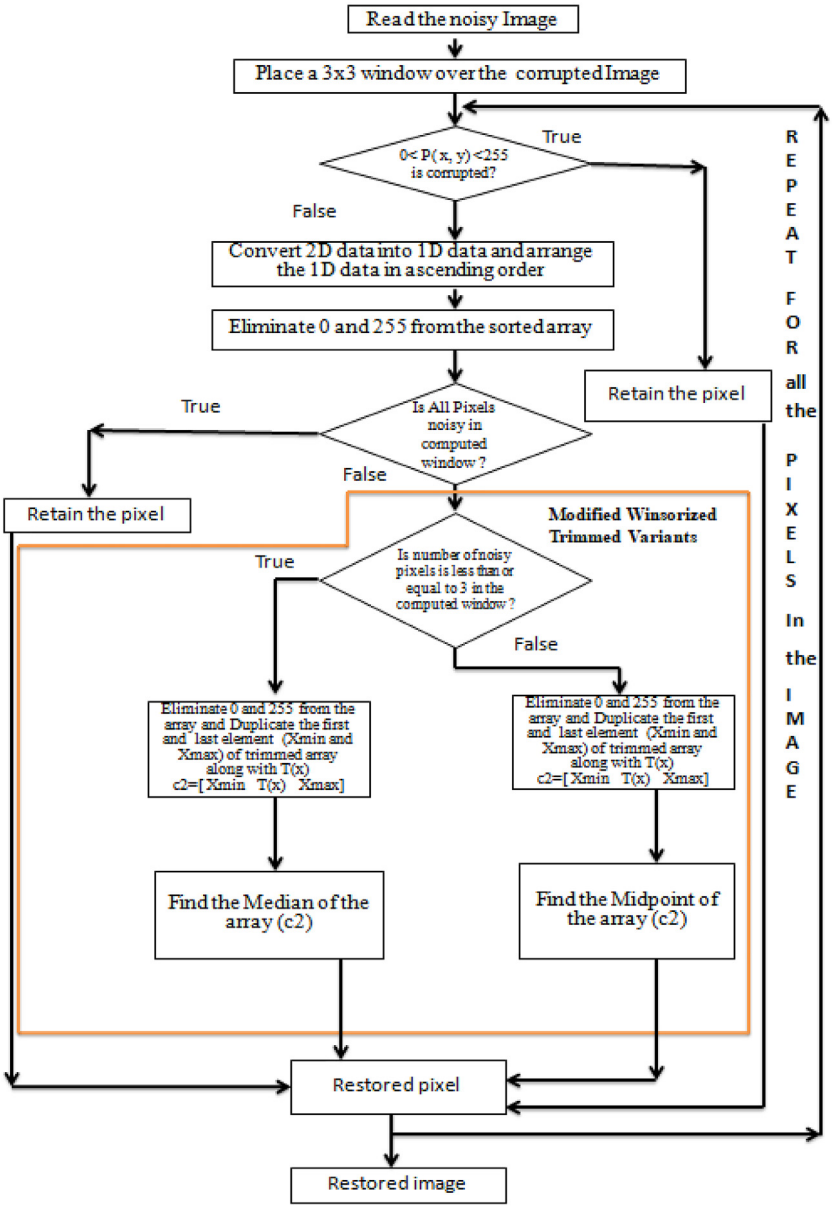


Fig. 1. Flowchart of Decision based Unsymmetrical Trimmed Modified Winsorized Variants (DBUTMWV).

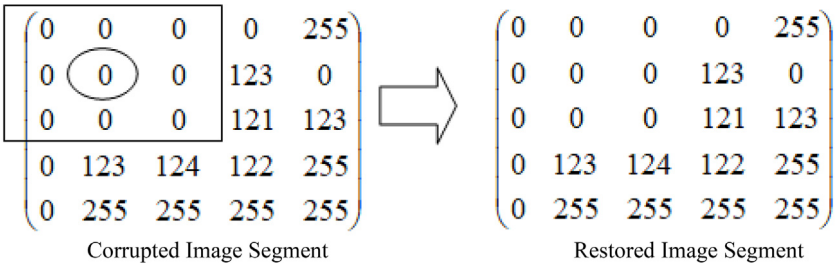


Fig. 2. Exhibit of case 1 in DBUTMWVF.

number of noisy pixels is 2. Convert 2D data into 1D data. Sort the data in increasing order. Trim the 0 and 255 in the array. Replace the corrupted pixel with Modified Winsorized median as illustrated in Fig. 5.

Modified Winsorized Median:

Replicate the minimum value and maximum value of the trimmed array and find the median of the resultant array.

Unsorted array: 124 128 255 126 255 121 126 124 121

Sorted Array: 121 121 124 123 126 126 128 128

Modified Winsorized Median: Median (121 121 121 124 124 126 126 128 128) = 124

Case (5): The Processed Pixel is not noisy

The Processed pixel holds 122 which is not an outlier. Hence the pixel in termed non noisy and left unaltered (see Fig. 6).

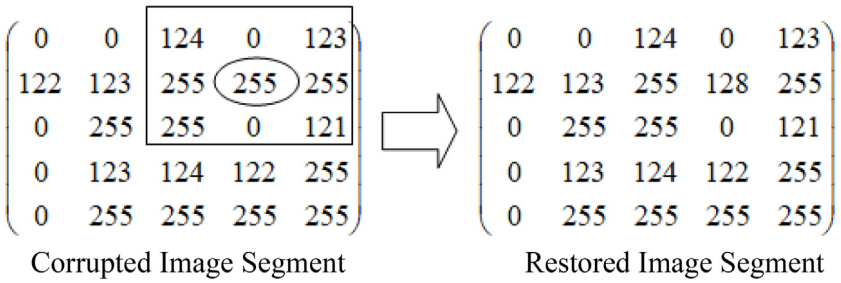


Fig. 3. Exhibit of case 2 in DBUTMWV.

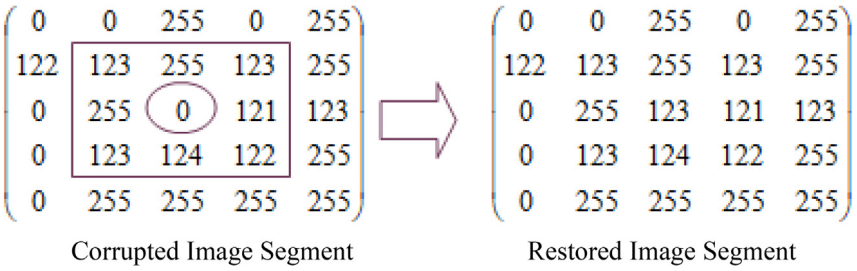


Fig. 4. Exhibit of case 3 in DBUTMWV.

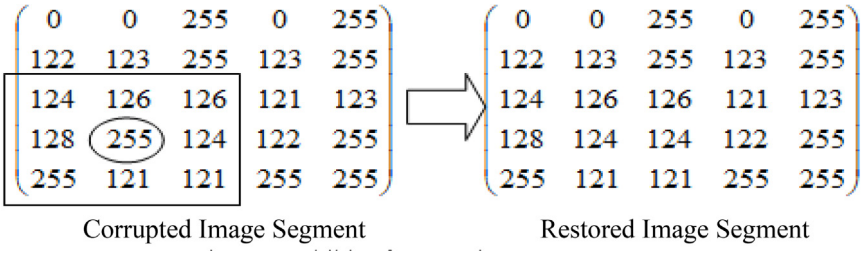


Fig. 5. Exhibit of case 4 in DBUTMWVF.

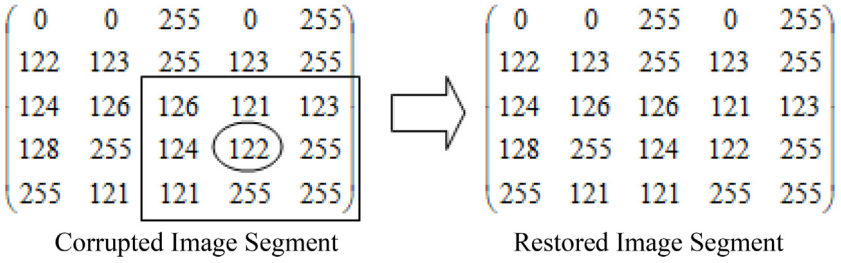


Fig. 6. Exhibit of case 5 in DBUTMWVF.

Table 1
Peak signal to noise ratio (PSNR) of different algorithms on Lena Image corrupted by salt and pepper noise.

Noise In %	SMF [1]	AMF [2]	PSMF [6]	αTMF α = 4 [15]	CWF [3]	DBA [10]	IDBA [11]	EDBA [20]	MDB MF [21]	CUT MPF [13]	MD BUTMF [22]	MDBU MF_GM [23]	ACBSA [16]	AWMF [24]	DB IDWIF [25]	ADKIF [18]	DBUT MWV(PA)
10	34.9	39.3	38.8	27.6	35.2	39	39.45	41.29	45.2	32.3	43.1	45.3	41.9	39.37	43.01	44.48	44.56
20	30.3	36.9	33.4	24.6	28.1	36.8	37.26	39.83	41.5	32.1	41.2	41.6	38.8	38.16	40.26	41.25	40.98
30	23.9	34.6	29.4	21	22.2	35.8	35.56	38.30	38.8	31.8	37.9	38.8	37.1	36.94	38.5	40.02	38.49
40	19	32.2	25.4	17.9	17.8	33.2	34.05	36.48	36.5	31.4	36.4	36.5	35.5	35.88	37.15	37.40	36.86
50	15.9	27.3	25.3	15.7	14.3	31.4	32.49	34.70	34.4	31.1	34.3	34.53	33.8	34.57	35.68	35.85	35.30
60	12.3	21.6	21.2	13.8	11.7	29.6	31.56	33.40	32.1	30.3	32.1	32.1	30.3	33.14	34.31	34.38	33.76
70	10	16.6	9.9	12.3	9.6	27.8	30.28	31.62	29.6	30.2	29.6	29.73	29.8	31.38	32.44	32.48	32.26
80	8.1	12.7	8.1	11.1	7.9	25.5	28.03	28.84	26.5	29.3	26.8	28.78	27.2	29.4	30.34	30.12	30.44
90	6.6	9.86	6.6	10.1	6.5	21.8	24.02	24.60	22.1	27.4	22.4	22.36	26.6	25.97	28.45	27.11	27.73

Table 2
Image Enhancement Factor (IEF) of different algorithms on Lena Image corrupted by salt and pepper noise.

Noise In %	SMF [1]	AMF [2]	PSMF [6]	αTMF α = 4 [15]	CWF [3]	DBA [10]	IDBA [11]	EDBA [20]	MDB MF [21]	CUT MPF [13]	MD BUTMF [22]	MDBU MF_GM [23]	ACBSA [16]	AWMF [24]	DB IDWIF [25]	ADKIF [18]	DBUT MWV(PA)
10	89.0	246.8	219.8	16.78	95.9	230.3	254.5	386.4	932	49.6	630.8	928	447	259.1	901.6	786.5	819.44
20	61.0	281.3	124.9	16.58	37.2	276.3	304	544.7	694.8	92.9	552.6	820	434	357.7	857.7	728.7	715.32
30	21.4	254.4	74.5	10.85	14.4	331.1	308	574.4	568.8	129.8	565.4	698	446	428.3	800.6	671.5	608.36
40	9.1	192.9	40.1	7.24	6.94	242.3	290.1	509.9	439.5	160.1	489.1	514	406	436.8	694.8	613.2	555.82
50	4.9	78.3	39.6	5.35	3.92	199.9	253.2	421.1	322.1	180.9	384.8	404	345	425.2	604.3	537.4	482.81
60	2.9	25	19.1	4.14	2.57	157.8	247.5	375.3	217	205.8	282.1	277	184	357.3	521.7	446.4	406.95
70	2.0	9.1	1.9	3.45	1.83	123.0	213.6	290.8	144.8	201.3	183.4	188	194	281.1	424.8	356.5	338.07
80	1.4	4.3	1.4	2.96	1.42	81.5	145.7	175.5	90.6	200.8	110.5	109	120	197.3	329.1	237.8	253.32
90	1.1	2.5	1.1	2.65	1.16	39.1	65	74.3	40.2	114.2	45.5	44	116	100.9	213.9	134.3	152.63

Table 3
Structural Similarity Index Metric (SSIM) of different algorithms on Lena Image corrupted by salt and pepper noise.

Noise In %	SMF [1]	AMF [2]	PSMF [6]	αTMF α = 4 [15]	CWF [3]	DBA [10]	IDBA [11]	EDBA [20]	MDB MF [21]	CUT MPF [13]	MD BUTMF [22]	MDBU MF_GM [23]	ACBSA [16]	AWMF [24]	DB IDWIF [25]	ADKIF [18]	DBUT MWV(PA)
10	0.931	0.981	0.98	0.869	0.932	0.970	0.97	0.97	0.992	0.895	0.992	0.922	0.987	0.979	0.998	0.991	0.991
20	0.881	0.973	0.94	0.627	0.837	0.962	0.96	0.97	0.983	0.893	0.982	0.983	0.975	0.971	0.980	0.983	0.983
30	0.718	0.958	0.88	0.369	0.609	0.950	0.95	0.96	0.971	0.888	0.971	0.972	0.963	0.962	0.971	0.970	0.970
40	0.445	0.928	0.76	0.206	0.340	0.930	0.93	0.95	0.955	0.881	0.957	0.957	0.948	0.952	0.961	0.960	0.958
50	0.216	0.835	0.55	0.124	0.155	0.903	0.91	0.94	0.931	0.872	0.938	0.938	0.927	0.938	0.946	0.944	0.944
60	0.093	0.607	0.09	0.078	0.067	0.866	0.89	0.92	0.897	0.862	0.910	0.910	0.866	0.919	0.929	0.923	0.925
70	0.041	0.300	0.04	0.050	0.033	0.814	0.86	0.89	0.846	0.85	0.870	0.870	0.850	0.892	0.902	0.899	0.901
80	0.018	0.110	0.02	0.033	0.016	0.735	0.81	0.83	0.764	0.831	0.800	0.803	0.769	0.85	0.861	0.857	0.867
90	0.009	0.041	0.01	0.021	0.009	0.592	0.68	0.69	0.60	0.789	0.676	0.673	0.749	0.773	0.811	0.798	0.808

Table 4
Pratts Figure of Merit (Pratts FOM) of different algorithms on Lena Image corrupted by salt and pepper noise.

ND in %	DBA [10]	IDBA [11]	MDBMF [21]	CUTMPF [13]	MD-BUTMF [22]	MDBU MF_GM [23]	ACBSA [16]	AWMF [24]	DBIDWIF [25]	ADKIF [18]	DBUT MWV(PA)
10	0.885	0.892	0.942	0.733	0.940	0.942	0.929	0.900	0.933	0.941	0.956
20	0.871	0.856	0.896	0.688	0.890	0.904	0.894	0.887	0.900	0.911	0.931
30	0.823	0.831	0.861	0.670	0.852	0.853	0.859	0.836	0.875	0.880	0.892
40	0.787	0.797	0.813	0.647	0.807	0.805	0.825	0.816	0.843	0.836	0.839
50	0.743	0.758	0.763	0.641	0.735	0.741	0.791	0.804	0.817	0.796	0.802
60	0.699	0.727	0.651	0.621	0.664	0.659	0.626	0.739	0.769	0.744	0.764
70	0.582	0.670	0.528	0.558	0.587	0.583	0.594	0.645	0.710	0.678	0.709
80	0.467	0.580	0.441	0.477	0.468	0.468	0.468	0.553	0.624	0.595	0.628
90	0.332	0.425	0.306	0.392	0.326	0.339	0.441	0.423	0.511	0.459	0.505

Table 5
Error Rate (ER) of different algorithms on Lena Image corrupted by salt and pepper noise.

ND in %	DBA [10]	IDBA [11]	MDBMF [21]	CUTMPF [13]	MDBUTMF [22]	MDBU MF_GM [23]	ACBSA [16]	AWMF [24]	DBIDWIF [25]	ADKIF [18]	DBUT MWV(PA)
10	20.85	21.01	8.78	21.34	8.84	8.9	8.72	10.39	8.29	8.23	8.55
20	24.75	24.86	17.5	25	17.53	17.67	17.36	18.35	16.77	16.75	17.36
30	30.5	30.64	26.6	31.13	26.69	26.42	26.19	26.28	25.37	25.24	26.76
40	37.75	37.57	35.81	38.96	35.75	35.68	35.12	34.54	33.9	33.94	36.55
50	45.7	45.91	45.24	48.74	45.27	45.22	43.66	43.08	42.51	42.65	46.34
60	53.96	54.49	55.02	60.04	55.16	55.12	53.57	52.07	51.3	51.3	56.93
70	63.08	63.31	65.63	71.6	65.55	65.7	62.48	61.31	60.35	60.23	67.75
80	72.83	72.56	76.93	83.1	77.05	76.98	71.47	71.21	69.61	70.04	78.68
90	83.29	82.46	88.97	93.1	87.97	89.02	81.38	82.32	79.65	80.37	89.61

4. Simulation results and discussions

The Performance of the Decision Based Unsymmetrical trimmed modified Winsorized variants is evaluated based on reference based Difference metric such as Peak Signal to noise Ratio (PSNR) as given in Eq. (1), Image Enhancement factor (IEF) as given in Eq. (3). The structure preservation after restoration is gauged by Structural Similarity Index Metric (SSIM) as illustrated in Eq. (5). The information preservation of the image is given by Pratt's FOM (Figure of Merit) as given in Eq. (6). The Error incurred while restoration is measured using Error rate as shown in Eq. (4). For a good noise removal algorithm, a high value of PSNR and IEF is very much required. The values of SSIM and FOM should be closer to 1 indicating better structure preservation and information retained. A low error rate indicate the impact of the algorithm while restoration. Apart from these measures Image Quality index metric (IDI), Blur Metric (BM), Normalized Cross Correlation Index metric (NCC) were used on different images and results were inferred.

$$\mathrm{PSNR} = 10\log_{10}\left(\frac{255^2}{MSE}\right) \tag{1}$$

$$\mathrm{MSE} = \frac{\sum_i \sum_j (k_{ij} - a_{ij})2}{rowXcol} \tag{2}$$

$$\mathrm{IEF} = \frac{\left(\sum_i \sum_j n_{ij} - r_{ij}\right)^2}{\left(\sum_i \sum_j x_{ij} - r_{ij}\right)^2} \tag{3}$$

Table 6
Performance of the DBUTMWV on different images of the database corrupted by 90% salt and pepper noise.

Images	PSNR	IEF	MSE	SSIM	FOM	Blur Metric	NCC	Error rate	UQI
aero.tiff	31.21	474	49	0.731	0.247	0.319	0.975	89.68	0.270
barbara.tif	22.06	41	404	0.626	0.486	0.373	0.974	89.83	0.458
bird.tif	26.34	109	150	0.841	0.515	0.453	0.996	89.54	0.476
elaine.png	25.49	90	183	0.667	0.438	0.429	0.990	89.83	0.473
fingerprint.bmp	18.08	16	1009	0.524	0.550	0.311	0.952	89.91	0.532
Hill.jpg	24.99	82	205	0.633	0.478	0.349	0.993	89.76	0.458
Butterfly.bmp	20.74	31	547	0.679	0.533	0.373	0.958	89.71	0.429
Pentagon.tif	24.40	65	235	0.580	0.494	0.336	0.990	89.85	0.391
two.bmp	21.49	34	460	0.663	0.529	0.358	0.968	89.59	0.457
Pirate.tiff	23.96	72	260	0.656	0.478	0.381	0.978	90.10	0.454
Zelda.png	28.3	179	96	0.805	0.467	0.462	0.990	89.79	0.626
House.bmp	23.36	55	299	0.731	0.474	0.370	0.993	89.53	0.357
Crater.png	26.44	103	147	0.613	0.450	0.363	0.993	89.78	0.367

Table 7
Peak signal to noise ratio (PSNR) of different algorithms on Baboon & Boat images corrupted by salt and pepper noise.

	IMAGE	Baboon	Boat	Baboon	Boat	Baboon	Boat	Baboon	Boat
	ND in %	30		50		70		90	
PSNR	AMF	26.9	30.21	24.7	27.31	22.4	24.08	19.58	20.37
	DBA	26.45	30.15	24.11	26.75	21	23.88	16.32	19.42
	MDMF	28.87	33	26.09	29.02	23.23	24.94	20.02	21.34
	MDBUTMF	28.41	29.67	27.08	26.15	24.07	21.71	19.24	18.81
	ACBSA	28.13	31.22	25.71	28.33	22.54	24.72	20.65	22.19
	AWMF	28.69	31.35	26.63	29.1	24.6	26.45	21.42	21.75
	DBIDWIF	29.79	33.77	26.98	30.45	24.74	27.86	22.12	24.33
	ADKIF	29.64	33.07	26.90	30.02	24.8	27.05	22.01	22.41
	DBUTMWV(PA)	29.28	32.90	26.68	29.6	24.69	26.93	22.07	22.75

Table 8
Structural Similarity Index Metric (SSIM) of different algorithms on Baboon & Boat images corrupted by salt and pepper noise.

	IMAGE	Baboon	Boat	Baboon	Boat	Baboon	Boat	Baboon	Boat
	ND in %	30		50		70		90	
SSIM	AMF	0.841	0.899	0.734	0.828	0.585	0.706	0.349	0.492
	DBA	0.84	0.89	0.73	0.82	0.52	0.71	0.28	0.5
	MDMF	0.9	0.94	0.8	0.87	0.64	0.74	0.34	0.48
	MDBUTMF	0.91	0.93	0.82	0.85	0.66	0.66	0.35	0.35
	ACBSA	0.88	0.92	0.8	0.85	0.58	0.72	0.37	0.56
	AWMF	0.887	0.926	0.806	0.875	0.685	0.786	0.452	0.589
	DBIDWIF	0.916	0.946	0.832	0.897	0.712	0.824	0.513	0.679
	ADKIF	0.906	0.944	0.819	0.89	0.7	0.806	0.482	0.614

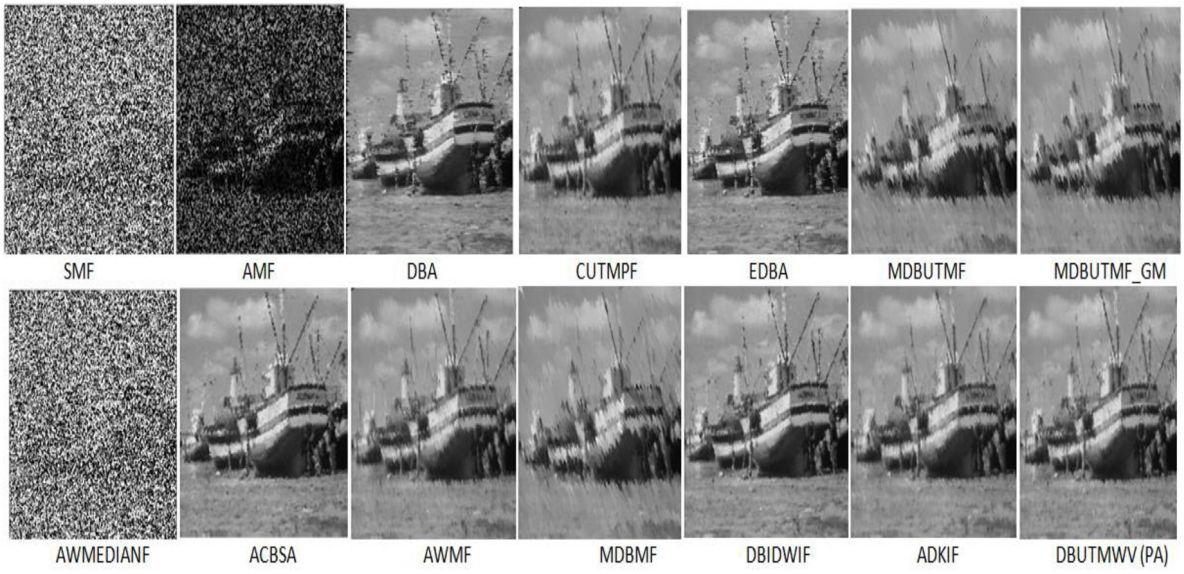


Fig. 7. Visual performances of algorithms on boat image damaged by 90% salt and pepper noise.

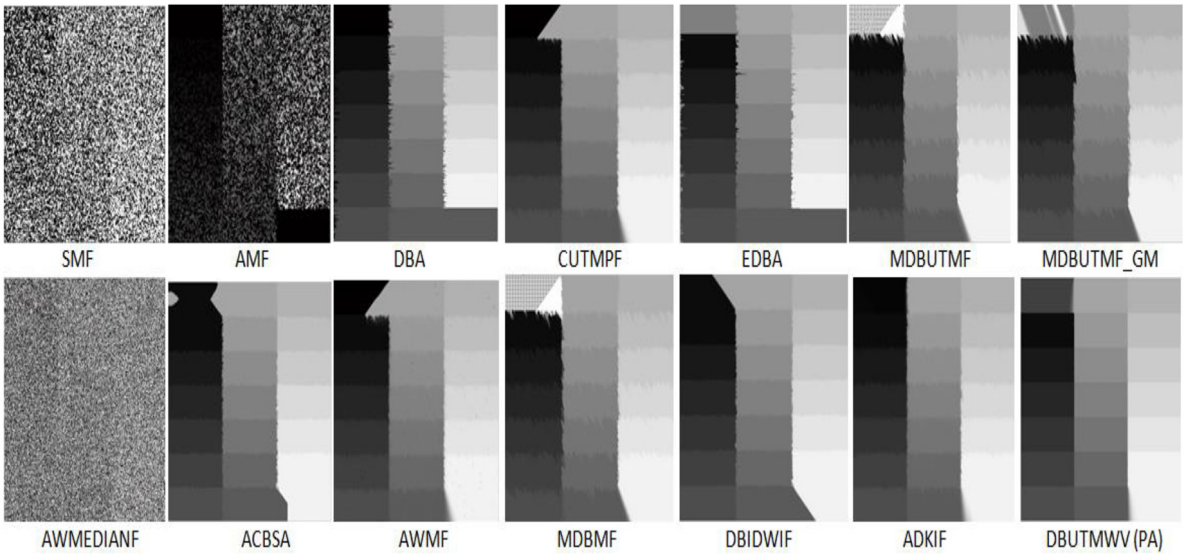


Fig. 8. Visual performances of algorithms on synthetic image damaged by 90% salt and pepper noise.

$$\text{Error Rate} = \frac{[\#R]}{row X col} X100\% \tag{4}$$

where a, g, k indicates Original image, corrupted image and restored image respectively. rowxcol gives the size of Processed image. R indicates to number of pixels that are restored is not equal to actual pixels in restored and original image respectively.

$$\text{SSIM}\ (a,k) = \frac{(2\mu a\mu k + Z1)\,(2\sigma ak + Z2)}{\left(\mu a^2 + \mu k^2 + Z1\right)\left(\sigma a^2 + \sigma k^2 + Z2\right)} \tag{5}$$

where a, k indicates original, restored image respectively, μa, μk gives the average of original, average of restored images respectively, σa, σk indicates the Standard deviation of original and restored image respectively. $Z1 = (F_1L)^2$, $Z2 = (F_2L)^2$ is used to stabilize weak denominator, L is the dynamic range of the pixel-values(for an 8 bit image it takes from 0 to 255) and the value F1 = 0.01 and F2 = 0.03 holds by default [26]. The information preservation performance of DBUTMWV is done using Pratt's figure of merit (Pratt's FOM) [27]. The FOM calculates the correlation between information content of two images, is given Eq. (6).

$$\text{Pratt's FOM} = \frac{1}{\text{MAX(Ya, Yk)}} \sum_{1}^{Yk} \frac{1}{(1+ \propto P^2)} \tag{6}$$

where Ya, Yk gives the information values of edges in the Original and restored image respectively. "P" is the distance between the information pixel of restored and original image respectively and the value of α is used as 1/9 referred from Pratt [19]. The image database used for experiments is obtained from University of southern California (signal and image processing institute) [28]. The algorithms used for comparison are derived from books and recent literatures. All the algorithms were run on these images and results were tabulated. Experiments were conducted by increasing the noise densities from 10% to 90% with increments of 10%. The filters that are used in this work are derived from few literatures indicated in square brackets. The Algorithms used for comparison were Progressive switched median filter(PSMF) [5], Decision based filter (DBA) [10], Enhanced Decision based filter (EDBA) [20], Modified Decision based Median filter (MDBMF) [21], Adaptive weighted median (AWMF) [24], Modified decision based unsymmetrical trimmed median filter (MDBUTMF) [22], Adaptive Decision based Kriging interpolation filter (ADKIF) [18], Adaptive cardinal B spline algorithm (ACBS) [16], Decision Based Inverse Distance weight interpolation filter (DBIDWIF) [25], Standard Median filter, Centre weighted median filter (CWF) [3], Adaptive median filter (AMF) [2], Modified Decision based unsymmetrical trimmed median filter with global mean (MDBUTMF_GM) [23], Alpha trimmed mean filter (αTMF) [1]. The experiments were done in 7nd generation i7-7500 CPU with an operating frequency of 2.70 GHz with a 16 GB RAM capability on MATLAB R2014a environment.

Tables 1, 2, 3, 4, 5 gives the Peak signal to noise ratio (PSNR), Image Enhancement Factor (IEF), Structural similarity Index metric (SSIM), Pratt's Figure of Merit (FOM) and Error Rate (ER) respectively of different algorithms on Lena image corrupted by 90% salt and pepper noise. It was found that the DBUTMWV algorithm has a higher PSNR and IEF values at increasing noise densities. This indicates the good noise removal characteristics of the algorithm. A higher value of IEF indicates that the improved quality of the image after removing the noise. The value of SSIM is closer to 1 indicating the structural preservation of the algorithm. The value of FOM is again closer to 1 indicating the data retaining capability of the algorithm. The wrongly detected pixel value is very less even at high noise densities. Hence the Error rate is better compared to other algorithms.

Algorithms such as SMF, AMF, PSMF, αTMF, CWF fails to remove noise at high noise densities. Hence these algorithms were not used for comparison in Tables 4 and 5. Table 6 gives the performance of DBUTMWV algorithm on different images of the database on 9 different quantitative measures. The performance of the DBUTMWV algorithm is further validated using Normalized cross correlation (NCC), Universal Image quality index metric (UIQI), Blur Metric (BM) along with the 5 measures used already. Different images of varying detail were subjected to the DBUTMWV algorithm which yielded different values of the quantitative measures. It was found that the DBUTMWV algorithm had excellent noise suppression characteristics with edge preservation. Tables 7 and 8 gives the quantitative performance such as PSNR, SSIM of different algorithms on varying noise densities on baboon and boat image. It was found that the DBUTMWV algorithm is good in eliminating high density noise by preserving the vital information of the image. Fig. 7 illustrates the qualitative performance of different algorithm on boat image corrupted by 90% salt and pepper noise. It was found visually that the algorithm fairs better than the other standard and existing algorithms. Fig. 8 is a synthetic image made of 21 visually differentiable grey levels. The variable grey levels can be distinguished as Step edge, roof edge and ramp edge. Step edge is the region between the first black block of first column and grey colour block in the second column. The horizontal variation of blocks is referred as ramp edge and the region between the last two white blocks of the last column is referred as roof edge. Different Algorithms were subjected to the synthetic image. It was found that SMF, AMF,

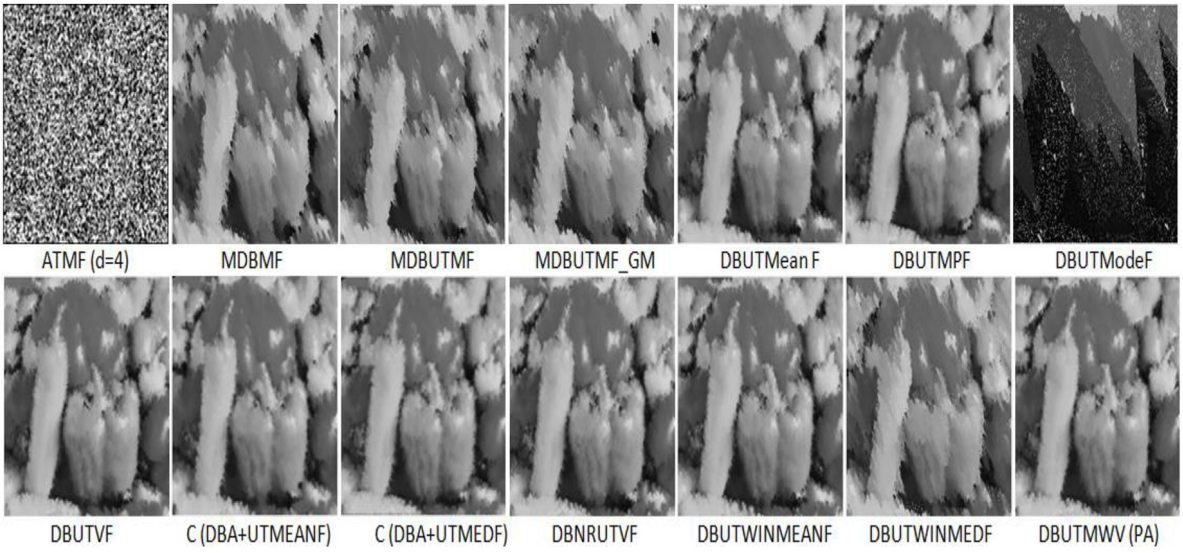


Fig. 9. Visual performances of trimmed filters on peppers image damaged by 90% Salt and pepper noise.

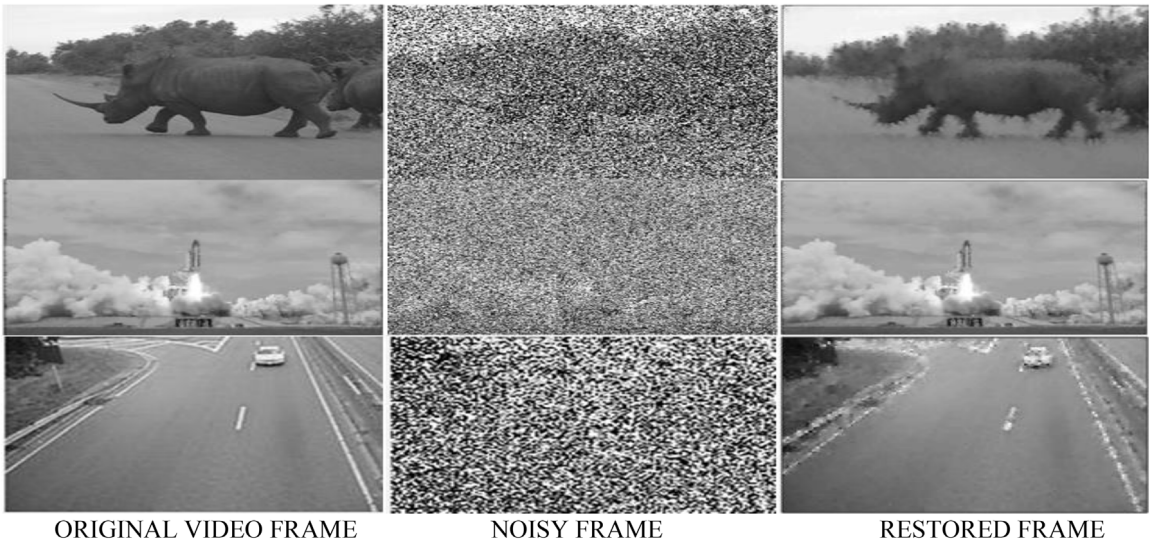


Fig. 10. Visual performance of the DBUTMWVF on rhinos.avi, shuttle.avi, viptraffic.avi corrupted by 50% salt and pepper noise.

AWMEDIANF fails to remove noise. Algorithms such as DBA, EDBA exhibits jittering of edges. Interpolation algorithms such as ACBSA, AKDIF, DBIDWF attenuates step edges are attenuated. These algorithms preserves ramp image. The DBUTMWV algorithm was found to exhibit very good results by preserving the ramp edge and step edge (slightly attenuated indicated in change in colour). The performance of the different trimmed filters is compared. Fig. 9 gives the performance of different trimmed filters on pepper image corrupted by 90% outliers. The algorithms that employed conventional alpha trimmed mean filter fails at high noise densities. Unsymmetrical trimmed statistics such as median, mean, midpoint attenuates the edges of the image. Modified Winsorized unsymmetrical statistic such as median and mean blurs the image. Cascaded trimmed filters (DBA + Trimmed filters) smudges the image. Visually the DBUTMWV algorithm looks good with information content highly visible after noise removal. Fig. 10 gives the performance of the DBUTMWV algorithm on videos such as rhinos.avi, shuttle.avi, viptraffic.avi corrupted by 50% salt and pepper noise. It was found that the DBUTMWV algorithm eliminates high density outliers in every frame and retains the information in each frame. At high noise densities the performance of the algorithm reduces and information content of the image is restored. The Algorithm used unsymmetrical trimmed modified Winsorized median or midpoint depending upon the number of noisy content of the processing window. The computed median or midpoint lies between conventional median and trimmed mean. This enables the statistic to perform better while removing high density outliers, Also the information content is well preserved.

5. Conclusion

The Decision based Unsymmetrical trimmed Modified Winsorized variants DBUTMWV is introduced for the removal of high density salt and pepper noise in images and videos. The algorithm exhibit excellent noise suppression characteristics with vital information of the images preserved. The performance of the algorithm was validated in terms of very good quantitative values and supporting qualitative aspects. These make the algorithm good for low level analysis.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Funding

This research did not receive any specific grant from funding agencies in the public, commercial or not for profit sectors.

References

[1] J. Astola, P. Kusmanen, Fundamentals of Non Linear Digital Filtering, CRC press, 1997.
[2] H. Hwang, R.A. Haddad, Adaptive median filters: new algorithms and results, IEEE Trans. Image Process. 4 (1995) 499–502.
[3] D.R.K. Brownrigg, The weighted median filter, Commun. ACM 27 (8) (1984) 807–818.
[4] S.J. Ko, Y.H. Lee, Center weighted median filters and their application to image enhancement, IEEE Trans. Circuits Syst. 38 (9) (1991) 984–993.
[5] Tao Chen, K. Ma, Tri-state median filter for image de-noising, IEEE Trans. Image Process. 8 (12) (1999) 1834–1837.
[6] Z. Wang, D. Zhang, Progressive switching median filter for removal of impulse noise from highly corrupted images, IEEE Trans. Circuits Syst.-II (46) (1999) 78–80.
[7] A. Hamza, H. Krim, Image denoising: a non linear robust statistical approach, IEEE Trans. Signal Process. 49-12 (2001) 304–305.
[8] H. Eng, K. Ma, Noise adaptive soft-switching median filter, IEEE Trans. Image Process. 10 (2) (2001) 242–251.
[9] Raymond H. Chan, Chung-Waa Ho, Mila Nikolova, Salt and pepper noise removal by median – type noise detectors and detail preserving regularization, IEEE Trans. Image Process. 14-10 (2009) 1479–1485.
[10] K.S. Srinivasan, D. Ebenezer, A new fast and efficient decision-based algorithm for removal of high-density Impulse noises, IEEE Signal Process. Lett. 14 (3) (2007) 189–192.
[11] S. Madhu, K. Nair, Rao Revathy, Tatavarti, An Improved Decision Based Algorithm for impulse noise removal, Proc. Congr. Image Signal Process. (2008) 426–431.
[12] S. Balasubramanian, S. Kalishwaran, R. Muthuraj, D. Ebenezer, V. Jayaraj, An efficient Non linear cascade filtering algorithm for removal of high density salt and pepper noise in image and video sequence, in: Proceedings in International Conference on Control, Automation, Communication and Energy Conservation, 2009, pp. 1–6.
[13] K. Vasanth, J. Tena, V. Elanangai, Amuthan, Cascaded algorithm for the removal of impulse noise variants and artifacts in images, Int. J. Appl. Eng. Res. 9 (19) (2014) 5779–5796.
[14] K. Vasanth, V. Jawahar Senthil Kumar, R. Rajesh, Anita, Cascaded decision based non linear filters for high density salt and pepper noise in images and videos, in: Proceedings in International Conference on Circuits, Power and Computing Technologies ICCPCT, 2013, pp. 855–860.
[15] Tian Bai, Jieqing Tan, Min Hu, Yan wang, A novel algorithm for removal of salt and pepper noise using continued fractions interpolation, Signal Process. 102 (2014) 247–255.
[16] P. Syamala Jayasree, Paru Raj, Pradeep Kumar, Rajesh Siddavatam, S.P. Ghrera, A fast novel algorithm for salt and pepper image noise cancellation using cardinal B-splines, Int. J. Signal Image Video Process. (2012) http://dx.doi.org/10.1007/s11760-012-0368-3, Springer.
[17] T. Veerakumar, R.P.K. Jagannath, B.N. Subudhi, et al., Circuits Syst. Signal Process. 36 (2017) 1192, http://dx.doi.org/10.1007/s00034-016-0352-1.
[18] Ramachandran Varatharajan, K. Vasanth, Gunasekaran Manogaran, M. Priyan, X.Z. Zao, An adaptive decision based kriging interpolation algorithm for the removal of highdensity salt and pepper noise in images, Comput. Electr. Eng. (2017) (2017) http://dx.doi.org/10.1016/j.compeleceng.2017.05.035.
[19] I.A. Abdou, W. Pratt, Quantitative design and evaluation of enhancement/thresholding edge detectors, Proc. IEEE 67 (5) (1979) 753–766.
[20] J. Tena, K. Vasanth, Govindaswamy Indhumathi, An enhanced decision based algorithm for the reduction of high density salt and pepper noise with reduced streaks, in: Proceedings in International Conference on Electronics and Communication Systems (ICECS -2014), 2014, Feb.13–14.
[21] K. Aiswarya, V. Jayaraj, D. Ebenezer, A New and efficient Algorithm for the removal of high density salt & pepper noise in images & videos, in: Proceedings in International Conference on Computer Modelling and Simulation, 2010, pp. 409–413.
[22] S. Esakkirajan, T. Veerakumar, Adabala N. Subramanyam, C.H. Prem Chand, Removal of high density salt and pepper noise through modified decision based Unsymmetrical trimmed median filter, IEEE Signal Process. Lett. 18 (5) (2011) 287–290.
[23] T. Veerakumar, S. Esakkirajan, Ila Vennila, An approach to minimize very high density salt and pepper noise through trimmed global mean, Int. J. Comput. Appl. 39 (12) (2012).
[24] Peixuan Zhang, Fang Li, A new adaptive weighted mean filter for removing salt-and-pepper noise, IEEE Signal Process. Lett. 21 (10) (2014) 1280–1283.
[25] V. Kishorebabu, G. Packyanathan, H. Kamatham, et al., J. Image Video Proc. 2017 (2017) 67, http://dx.doi.org/10.1186/s13640-017-0215-0.
[26] Z. Wang, A.C. Bovik, H.R. Sheikh, E.P. Simoncelli, Image quality assessment: From error measurement to structural similarity, IEEE Trans. Image Process. 13 (4) (2004) 102–109.
[27] I.A. Abdou, W. Pratt, Quantitative design and evaluation of enhancement/thresholding edge detectors, Proc. IEEE 67 (5) (1979) 753–766, 1979.
[28] http://sipi.usc.edu/database/ (Accessed on 5 Februaray 2018).